



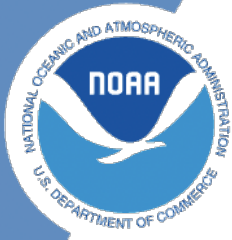
NOAA Technical Memorandum NMFS-XXX-##

NOAA quarto book

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1 Introduction

Here we will be displaying various data summaries and analyses related to dolphin management in the U.S. South Atlantic.

1.1 Load Libraries

The documentation of this data exploration was developed in R, using R Studio and Quarto. The following R libraries were also used. The `pals` package allows for improved visualization with different color palettes (**pals?**). The `rerddap` package allows for efficient downloads of physical data from the ERDDAP server (**rerddap?**).

2 Chapter 1 - NOAA Fisheries landings data

Here we will explore commercial and recreational landings data accessed through the FOSS (Fisheries One Stop Shop) data portal. The advantage of this portal is that commercial and recreational data are available side-by-side in the same units (pounds landed).

2.1 Data access and upload

Data are downloaded from the NOAA Fisheries [FOSS site] (<https://www.fisheries.noaa.gov/foss/f?p=215:200:660844904940::NO:RP::>). For the Data Set, we choose both commercial and recreational. For the Years, we select all years from 1980 to 2021. For NMFS Region, we select the "NMFS Regions" button and then select the Gulf, South Atlantic, Middle Atlantic and New England from the box. For the Species, we select "Dolphinfish" and "Dolphinfish**." For Report Format, we choose TOTALS BY YEAR/REGION. Hitting the "RUN REPORT" button then generates the data needed for the subsequent analysis, which can be downloaded under the "Actions" button.

We then read in the data file and clean it for analysis (removing the commas from the numbers so that R recognizes the values as numbers). We output the file header to make sure it was read in and that the numbers were converted correctly.

```
# clear workspace
rm(list = ls())

# read in data file
d <- read.table("data/foss_landings_comrec_USEEZ.csv", header = T, sep = ",")
head(d)
```

	Year	Region.Name	Pounds	Dollars	Collection	Metric.Tons
1	2000	Gulf	582,576	689,637	Commercial	264
2	2000	Gulf	5,641,491		Recreational	2,559

2 Chapter 1 - NOAA Fisheries landings data

```
3 2000      New England      6,545 12,414  Commercial      3
4 2000  South Atlantic    537,077 821,743  Commercial     244
5 2000  South Atlantic 16,049,049      Recreational    7,280
6 2000  Middle Atlantic    36,048 65,004  Commercial     16
```

```
# take commas out of file -----
d$Pounds <- as.numeric(gsub(",", "", d$Pounds))
d$Dollars <- as.numeric(gsub(",", "", d$Dollars))
d$Metric.Tons <- as.numeric(gsub(",", "", d$Metric.Tons))
head(d)
```

	Year	Region.Name	Pounds	Dollars	Collection	Metric.Tons
1	2000	Gulf	582576	689637	Commercial	264
2	2000	Gulf	5641491	NA	Recreational	2559
3	2000	New England	6545	12414	Commercial	3
4	2000	South Atlantic	537077	821743	Commercial	244
5	2000	South Atlantic	16049049	NA	Recreational	7280
6	2000	Middle Atlantic	36048	65004	Commercial	16

2.2 Summarize the landings

We then summarize the landings (in pounds) by year, sector, and region. The result is a new data frame with each row representing a year and each column representing the total landings for each region and sector.

```
# compile by year for sectors and regions -----
d$Collection <- factor(d$Collection, levels = c("Recreational", "Commercial"))
tab <- tapply(d$Pounds, list(d$Year, d$Collection, d$Region.Name), sum, na.rm = T)
tab[is.na(tab)] <- 0
#head(tab)

dat <- data.frame(cbind(tab[, , "Gulf"], tab[, , "South Atlantic"],
                        tab[, , "Middle Atlantic"], tab[, , "New England"]))
names(dat) <- c("rec_Gulf", "com_Gulf", "rec_SA", "com_SA", "rec_MA", "com_MA", "rec_NE", "com_NE")
head(dat)
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
1980	0	94984	0	76042	0	400	0	200

2 Chapter 1 - NOAA Fisheries landings data

1981	1559544	73075	8421898	59037	0	0	0	0
1982	4197503	162457	7573035	143960	2097	200	0	0
1983	2162936	190808	9134022	127986	45927	1400	0	400
1984	1189573	282976	7265430	158901	0	1700	0	400
1985	2542089	269477	6684821	142684	113666	5000	0	4800

2.3 Plot the summary

Now we plot the data, using similar color schemes by region. Because the recreational data have high imprecision prior to the 1990s we plot from 1990 to the last year of data.

```
# select years -----
dat <- dat[which(rownames(dat) > 1990), ]

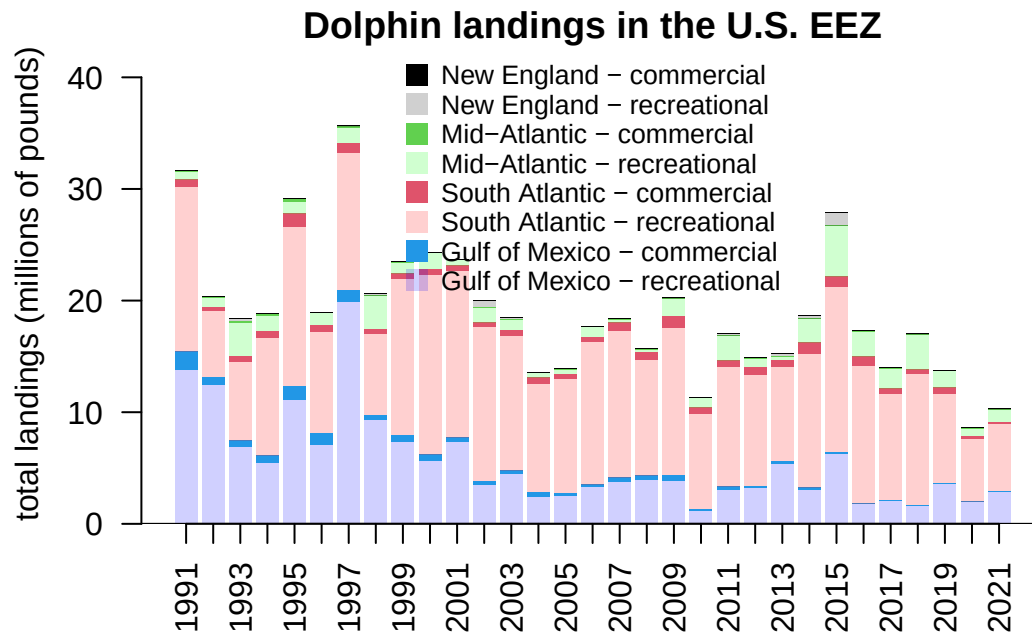
# define colors and plot -----
cols <- c("#0000FF30", 4, "#FF000030", 2, "#00FF0030", 3, "#00000030", 1)

par(mar = c(4, 4, 1, 0), mgp = c(2.5, 1, 0))
yrs <- as.numeric(rownames(dat))
#yrs[seq(2, 32, 2)] <- NA

b <- barplot(t(dat)/10^6, names.arg = yrs, col = cols, ylim = c(0, 43),
             border = NA, las = 2, ylab = "total landings (millions of pounds)",
             main = "Dolphin landings in the U.S. EEZ")

#mtext(side = 3, line = -1.15, "Source: NMFS commerical and recreational landings statistic

legend("top", c("New England - commercial", "New England - recreational",
               "Mid-Atlantic - commercial", "Mid-Atlantic - recreational",
               "South Atlantic - commercial", "South Atlantic - recreational",
               "Gulf of Mexico - commercial", "Gulf of Mexico - recreational"),
      col = cols[8:1], pch = 15, pt.cex = 1.5, bty = "n", y.intersp = 0.9, cex = 0.85)
#legend(15, 30, c("Commercial", "Recreational"), col = c(1, "#00000060"),
#      pch = 15, pt.cex = 2, bty = "n", y.intersp = 1.1)
axis(1, at = b, lab = rep("", length(b)), pos = 0)
abline(h=0)
```



3 Updated Gulf/Atlantic parsing of data

There are some issues with the way that the FOSS system splits out Gulf and Atlantic catches by Monroe County, which is on the border between the two jurisdictions. We received data from Mike Larkin of SERO which has the correct split out of data for Monroe County between the Gulf and Atlantic. We compare the total recreational catch for the Gulf and Atlantic between the FOSS data and the updated data. The numbers are very close; the average difference from the FOSS data is 2%.

```
ml <- read.table("data/RecreationalDolphinLandings_SAandGOM_MLarkin.csv", header = T, sep = ",")
ml <- ml[which(ml$Year > 1990), ]

# select same years for FOSS data set
dat <- dat[which(yrs <= max(ml$Year)), ]

# check that years match
summary(rownames(dat) == ml$Year)
```

```
Mode      TRUE
logical    31
```

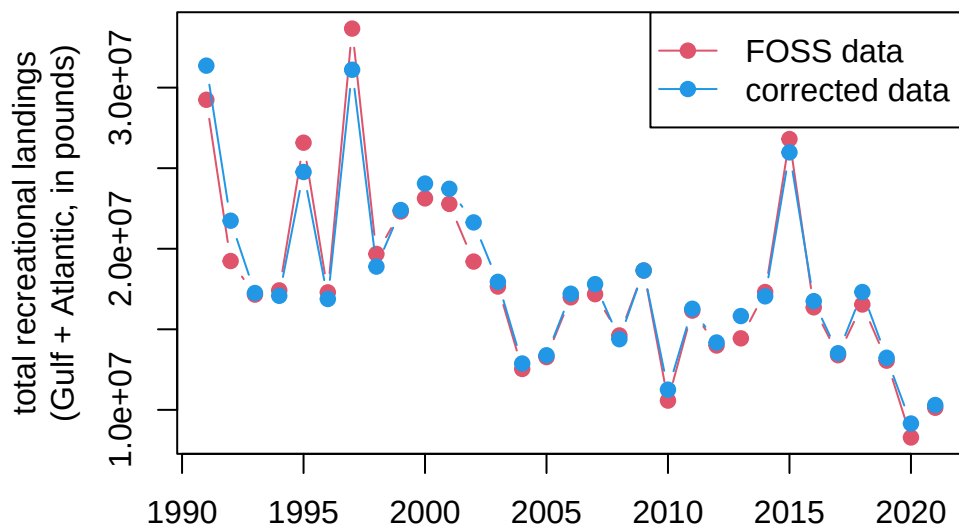
```
yrs <- as.numeric(rownames(dat))

# check that totals match
old <- dat$rec_Gulf + dat$rec_SA + dat$rec_MA + dat$rec_NE
new <- ml$GOM + ml$ATL

# plot totals
par(mar = c(4, 6, 2, 1), mgp = c(2.5, 1, 0))
matplot(yrs, cbind(old, new), xlab = "", lty = 1, col = c(2, 4), type = "b", pch = 19,
        ylab = "total recreational landings\n(Gulf + Atlantic, in pounds)",
        main = "Comparison of FOSS data with corrected split by Monroe County")
legend("topright", c("FOSS data", "corrected data"), col = c(2, 4), pch = 19, lty = 1)
```

3 Updated Gulf/Atlantic parsing of data

Comparison of FOSS data with corrected split by Monroe



```
mean((old - new)/old*100)
```

```
[1] -2.023886
```

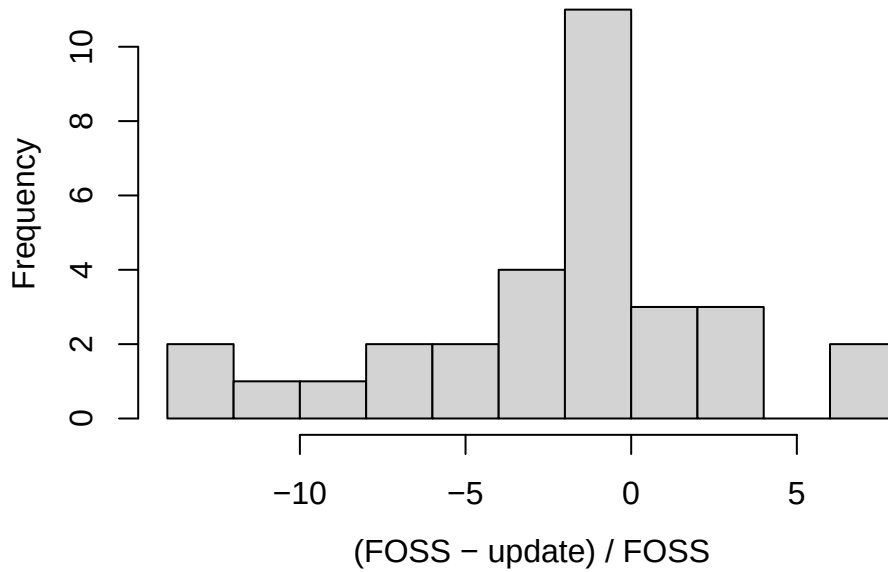
```
range((old - new)/old*100)
```

```
[1] -13.008552  7.587789
```

```
hist((old - new)/old*100, main = "Percent difference from FOSS data", breaks = 10,  
     xlab = "(FOSS - update) / FOSS")
```

3 Updated Gulf/Atlantic parsing of data

Percent difference from FOSS data



```
# replace FOSS data with update data
datnew <- dat
datnew$rec_Gulf <- ml$GOM
datnew$rec_SA <- ml$ATL - (datnew$rec_MA + datnew$rec_NE)
tail(dat)
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
2016	1777348	37733	12357817	828435	2212513	61336	16499	15747
2017	2064106	68260	9457460	533082	1820541	16179	48607	30969
2018	1626095	57052	11713866	435290	3104018	20514	97107	12667
2019	3567170	85077	7967978	563708	1518629	47981	66	16918
2020	1952241	26653	5653294	260503	679556	14040	0	1880
2021	2893817	27212	6061024	177263	1051785	25799	123038	11421

```
tail(datnew)
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
2016	751499	37733	13766851	828435	2212513	61336	16499	15747
2017	868489	68260	10777104	533082	1820541	16179	48607	30969
2018	508570	57052	13603875	435290	3104018	20514	97107	12667

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2019	1304865	85077	10410599	563708	1518629	47981	66	16918
2020	555695	26653	7916891	260503	679556	14040	0	1880
2021	1174412	27212	7950716	177263	1051785	25799	123038	11421

We replace the FOSS values for Gulf and South Atlantic recreational catches with these new values and replot the full data set.

